

COSC 39: Theory of Computation

Problem 1. Binary addition.

Let's leave the theory land and apply our knowledge in finite automata to some real problems.

For any string $w \in \{0, 1\}^*$, let $\langle w \rangle_2$ denote the integer represented by w in binary. For example:

$$\langle \epsilon \rangle_2 = 0, \quad \langle 0011 \rangle_2 = 3, \quad \langle 1010 \rangle_2 = 10, \quad \langle 1111 \rangle_2 = 15.$$

Let L and L' be arbitrary automatic languages over the alphabet $\{0, 1\}$. Prove that the following language is also automatic:

$$\{w \in \{0, 1\}^* : \langle w \rangle_2 = \langle x \rangle_2 + \langle y \rangle_2 \text{ for some } x \in L, y \in L'\}.$$

Hint: Try to prove first the case when both L and L' are languages containing a single binary integer.

Solution.

Problem 2. When does an NFA accept everything?.

Let N be an arbitrary NFA with n states for some alphabet Σ of size at least 2. How long can the shortest word rejected by N be?

Put differently, let $L(N)$ be the language recognized by the NFA N , and define

$$\Sigma^{\leq f(n)} := \Sigma^0 \cup \dots \cup \Sigma^{f(n)}.$$

What is the smallest $f(n)$ such that

$$\Sigma^{\leq f(n)} \subseteq L(N) \implies L(N) = \Sigma^*?$$

(a) Prove that $f(n) \leq 2^n$.

(b) Consider the alphabet

$$\Sigma := \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \# \right\}.$$

Define s_n to be the following word:

$$s_n := \# \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} \dots \begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{n \text{ characters}} \# \begin{bmatrix} 0 \\ 0 \end{bmatrix} \dots \begin{bmatrix} 1 \\ 0 \end{bmatrix} \# \dots \# \begin{bmatrix} 1 \\ 1 \end{bmatrix} \dots \begin{bmatrix} 0 \\ 1 \end{bmatrix} \#.$$

Notice that if we drop the brackets, then we can interpret the 0s and 1s between two consecutive # symbols (referred to as a *chunk*) as two binary numbers stacked on top of each other.

If we read the top numbers in every chunk from left to right, they form a binary counter from 0 to $2^n - 1$. For the bottom numbers, they form a binary counter from 1 to 2^n .

For example,

$$s_2 := \# \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \# \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \# \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \#$$

the top numbers are 00, 01, 10, and the bottom numbers are 01, 10, 11.

Construct an NFA recognizing the language $\Sigma^* \setminus \{s_n\}$.

Hint: You need to build separate gadgets to check that each chunk between two consecutive # has length exactly n , the two counts within a chunk are differed by one, the counts in adjacent chunks are consistent. For full credit, you can use at most $C \cdot n + o(n)$ states for constant C . Now you just proved $f(n) \geq \Omega(2^n/C)$.

★ (c) Prove or disprove that $f(n) \geq 2^{n-o(n)}$.

Solution.

- (a) Let N be an arbitrary NFA with n states for some alphabet Σ of size at least 2. Then we know that there exists some DFA D that is equivalent to this NFA, restricting to reachable states we also know that D has at most 2^n states.

Suppose $L(N) \neq \Sigma^*$. Let w be the shortest word rejected by D . Thus there exists a walk through states $q_0, q_1, \dots, q_r$ such that q_r is not an accepting state. Now note that since DFAs are deterministic, if a state repeats, i.e., $q_i = q_j$ for some $i \neq j$ then the (shorter) word obtained by deleting $w[i+1] \dots w[j]$ also ends in a non-accepting state, contradicting our assumption. Hence, $|w|$ is at most the length of the longest path in D . Since we don't visit a state more than once, this length is at most the number of states in D minus 1 (starting state), i.e.,

$$|w| \leq 2^n - 1$$

Thus, if all words of length $2^n - 1$ are accepted, then no words are rejected and $L(N) = \Sigma^*$. Therefore,

$$f(n) \leq 2^n - 1 \leq 2^n$$



